Appendix A: Power Analysis

We use actual Denver PD data to conduct a simulated power analysis. Using reported crime events from January 1st 2021 through April 30th 2024, we conduct a hypothetical intervention and examine downstream impacts on violent crimes. The simulation is as follows – every week, we take the top 100 offenders (according to the counts of prior violent and gun crimes). Out of those top 100, we randomly select 5 individuals to receive the treatment. The remaining 95 are control cases.

Subsequent weeks we conduct this same process again, but only include in the top 100 individuals who have not previously received the treatment. Thus, this RCT is similar to a stepped wedge design, (controls can later turn into treated) although we do not have a full population from the start (as new individuals can enter into the study at later dates).

We can then build a panel dataset of individuals counts of crime per week. We then estimate a Poisson model:

$$\log(\mathbb{E}[V_{it}]) = \beta_0 + \beta_1 \text{treat}_{it} + \gamma_t$$

Where $\text{treat\_}it$ is equal to 1 in the weeks after the treatment occurred for individual i, and zero otherwise. γ_t is a week fixed effect, to take into account potential seasonality of violent crimes over time. And we estimate this Poisson regression equation using cluster robust standard errors at the person level.

We start the treatment after 24 weeks (so there is historical data to rank offenders by). In practice we will fit a machine learning model on historical data to prioritize individuals, but for power analysis a simple ranking should be sufficient to illustrate whether our implementation is effective.

In the simulation, we multiple the observed number of violent and weapon offenses in the treated group by multiples of 0 (no effect), 0.9 (small effect), 0.8 (middle effect), and 0.5 (large effect). So the large effect would be on average reducing 50% of crimes in the treated group, the small effect would be reducing crimes by 10%, etc.

The plot below shows this simulation using the Denver data. Replication code can be found in the *power.py* python file. The code and data to replicate can be downloaded at https://www.dropbox.com/scl/fi/3448c35wgognxlpusn3xv/DPD_Data.zip?rlkey=ycgab6q6rhuskpqrbtzwusn2c&dl=0.

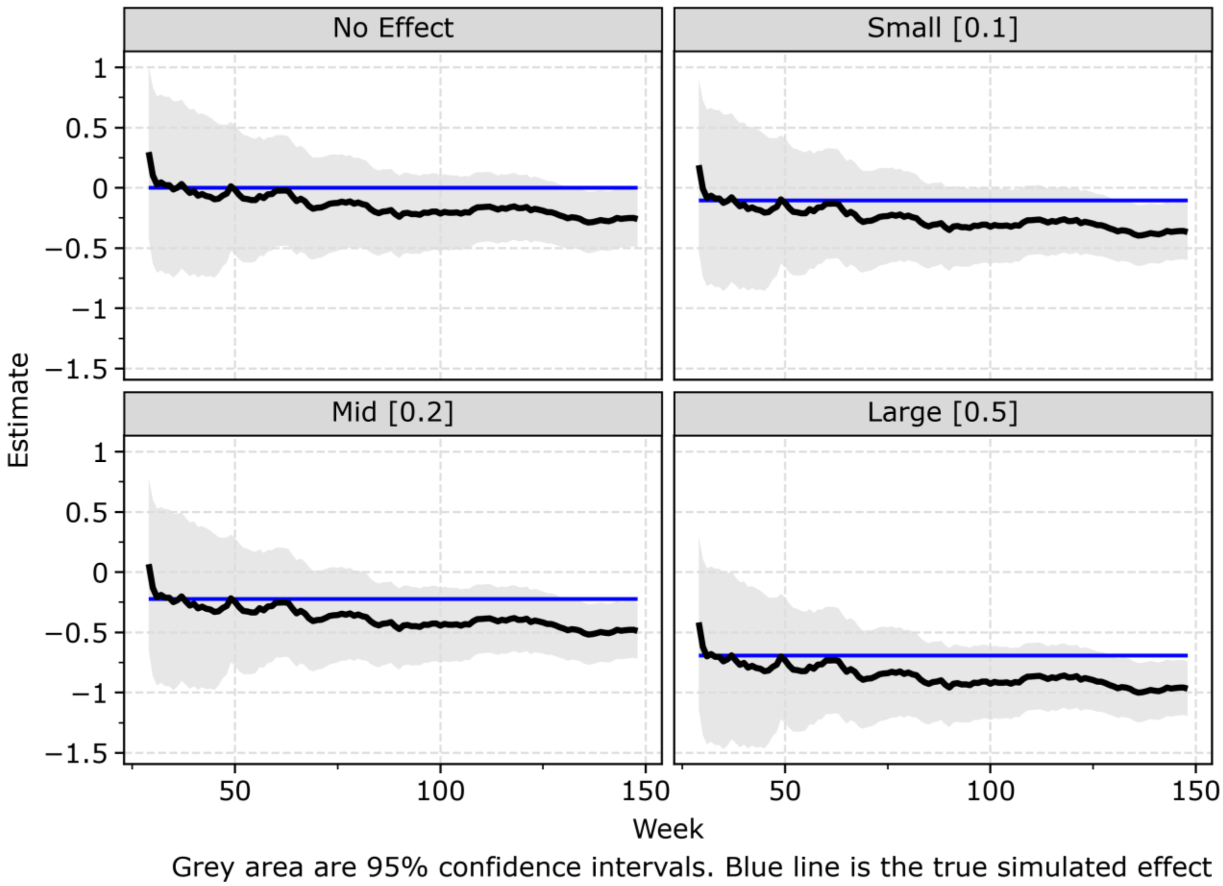

Figure A1: Plot showing simulated effects for the treatment over three years.

For the no effect condition, there is a slight bias for crime reduction. It is possible that the design, due to selecting high risk individuals, has a regression to the mean effect. Although the 95% confidence interval covers 0 until more than two years into the study. The large effect, a 50% reduction, according to this simulation would need to run for approximately a year before the confidence interval around the logged incident rate ratio from the Poisson regression equation does not cover 0. At two years, for each of the estimates, the standard error of the logged incident rate ratio is around 0.15, and at the end of our study window (a few weeks short of three years) the standard error is 0.12.

Based on this, we plan to monitor the experimental findings for 1 to 3 years. A 20% reduction is not unrealistic if the program is effective. Although a 50% reduction seems quite optimistic.